STRATEGIC ENERGY INFRASTRUCTURE PROPOSAL — REPUBLIC OF ESTONIA

Estonia Integrated Green Hydrogen Energy System

A Proposal for a Nationally Integrated Solar–Hydrogen Energy Production, Storage and Export Framework Aligned with ENMAK 2035 and the EU REPowerEU Mandate

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Date: March 2026

Classification: Public - open for review and discussion

Intended for: Ministry of Climate, Elering AS, Enterprise Estonia, EU Innovation Fund, Estonian Association of Engineers

This document is an independent strategic proposal. All figures are engineering estimates based on publicly available data. A full feasibility study is recommended prior to any investment decision.

EXECUTIVE SUMMARY

Estonia faces an acute and worsening energy paradox. The scheduled closure of its oil shale power plants — the backbone of national electricity supply for over a century — is creating a structural electricity deficit precisely as global energy prices surge and regional grid interconnection costs compound. The Baltic states currently import approximately 40% of their electricity, at an average cost 60% higher than the price they receive for exports. Between 2019 and 2026 alone, Estonian electricity prices rose by 74.3%, placing significant strain on households and industrial competitiveness alike.

This proposal sets out a technically grounded, financially structured and politically timely response: the **Estonia Integrated Green Hydrogen Energy System (EIGHES)** — a seasonal solar-to-hydrogen production, storage and export framework that would simultaneously address Estonia's winter energy gap, generate a nationally significant export revenue stream, reduce dependency on imported electricity, and position Estonia as a first-mover green hydrogen supplier within the European Union.

The system in brief: During summer months (May–September), a large-scale coastal solar installation powers an industrial electrolysis facility, fed by a Baltic Sea desalination plant taking advantage of the sea's uniquely low salinity. The hydrogen produced is stored in underground geological caverns or pressurised surface tanks. During winter (November–March), stored hydrogen is reconverted to electricity through combined-cycle turbines and fuel cells, partially filling Estonia's seasonal generation deficit. Crucially, a significant proportion of hydrogen is *not* reconverted but exported directly via pipeline to Germany, Finland and the broader EU hydrogen backbone network, where industrial demand is rapidly growing and prices are substantially higher than the equivalent electricity value.

Key headline figures:

Summary of Key Parameters

Total capital investment: 8.5–16 billion

Winter electricity delivered: 1.5–2.0 TWh/year

Solar capacity required: 10,000–14,000 MW

Desalination capacity: 6,000–8,000 m^3/day

Target hydrogen output: 300,000–500,000 t/year

Hydrogen export revenue: 0.8–1.6 bn/year

Payback period (w/ EU subsidy): 18–25 years

Jobs created (construction): 8,000–15,000

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1. Problem Statement: Estonia's Energy Crisis

1.1. Historical Context

For most of the twentieth century, Estonia enjoyed an enviable position among European nations: near-total energy self-sufficiency, powered by vast deposits of oil shale concentrated in the northeastern Ida-Viru county. At peak operation, the Narva power plants provided close to 90% of the country's electricity from domestically mined fuel. This self-reliance insulated Estonia from the energy price shocks that repeatedly destabilised its neighbours.

That era is ending. Under binding commitments made to the European Union, Estonia has committed to ending oil shale electricity production by 2035 and all oil shale energy use by 2040. The environmental rationale is sound — oil shale combustion is among the most carbon-intensive energy processes industrially operated in Europe — but the transition is creating a structural void that current planning does not adequately address.

1.2. The Import Dependency Problem

The Baltic states collectively import approximately **40% of their electricity**, making them among the most import-dependent electricity markets in the EU. Estonia's position within this picture is striking:

- Estonia imports electricity at an average price of **94.5/MWh**
- Estonia exports electricity at an average price of only **59.3/MWh**
- This structural buy-high, sell-low dynamic arises because Estonia imports during peak winter demand periods when Nordic-Baltic grid prices are highest, and exports marginal summer surplus when prices are lowest
- Between 2019 and 2026, the average electricity price paid by Estonian consumers rose by **74.3%** — from 88.7/MWh to 154.6/MWh
- The economic cost of this price increase amounts to approximately **512 million per year** of additional burden on the Estonian economy

1.3. The Synchronous Area Transition

In 2025, Estonia, Latvia and Lithuania completed their historic disconnection from the Soviet-era BRELL ring (the synchronised grid shared with Russia and Belarus) and synchronised with the Continental European grid. While this is strategically essential for energy security and geopolitical independence, it introduces new technical obligations: the Baltic states must now internally provide the frequency regulation services and spinning reserves that were previously shared across a much larger system. This requires additional controllable generation capacity that is currently

lacking.

1.4. The Seasonal Storage Gap

The central technical challenge of the energy transition in the Nordic-Baltic region is one of seasonal mismatch. Solar power peaks in June and July; wind power is moderate year-round but strongest in autumn and winter; electricity demand peaks sharply in December and January. Battery technology, even at its most advanced, addresses storage cycles of two to eight hours. No commercially available technology currently bridges the gap between a summer solar surplus and a winter electricity deficit — this is the fundamental problem that this proposal directly addresses.

The Core Problem in One Sentence

Estonia is simultaneously closing its only source of firm domestic electricity, accumulating renewable capacity that produces most in summer, and facing its largest electricity demand in winter — with no approved mechanism to bridge the seasonal gap.

2. The Proposed Solution: EIGHES

2.1. System Overview

The Estonia Integrated Green Hydrogen Energy System (EIGHES) proposes to resolve the seasonal storage problem not by attempting to store electricity directly, but by converting summer solar electricity into a *chemical energy carrier* — hydrogen — that can be stored indefinitely, transported by pipeline, and reconverted to electricity or sold industrially at high value.

The system comprises five integrated components operating in seasonal phases:

1. **Coastal Solar Array** — Large-scale photovoltaic installation on the Estonian coast and western islands
2. **Baltic Sea Desalination Plant** — Provides purified water feedstock for electrolysis, exploiting the Baltic's uniquely low salinity
3. **Industrial Electrolysis Facility** — Converts solar electricity and purified water into green hydrogen at scale
4. **Geological Hydrogen Storage** — Underground or surface storage of compressed hydrogen for seasonal carry-over
5. **Winter Power Generation & Export Hub** — Hydrogen-to-electricity conversion for domestic supply, plus pipeline export to EU hydrogen network

2.2. System Flow Diagram

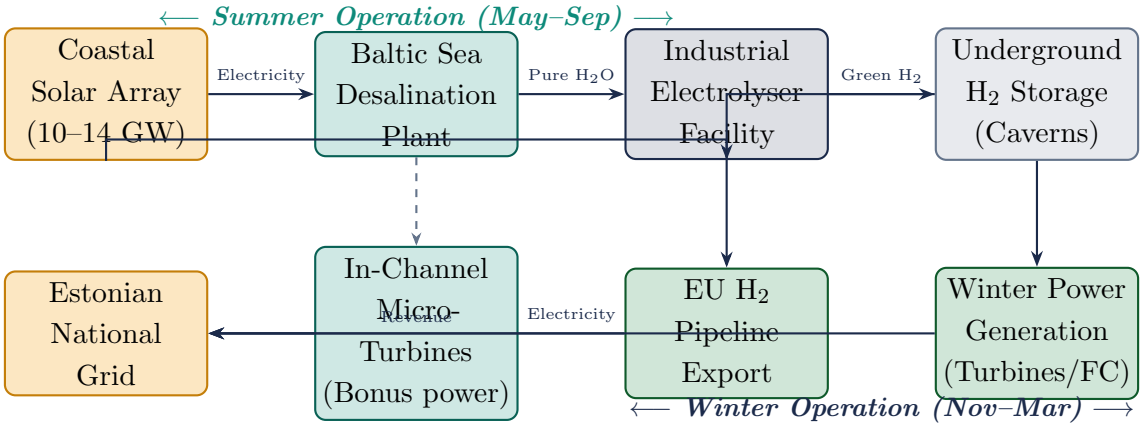


Figure 1: EIGHES integrated system flow. Summer solar electricity drives desalination and electrolysis; stored hydrogen powers winter electricity generation and EU export.

3. Technical Specification

3.1. Component 1: Coastal Solar Array

3.1.1. Site Selection

The solar installation would be situated primarily across three complementary locations:

- **Western Estonian coastline and islands (Saaremaa, Hiiumaa, Vormsi):** These offer the best solar irradiance in Estonia, prevailing onshore winds to keep panels cooled and efficient, proximity to the sea for the desalination feedwater pipeline, and relatively low land value compared to the mainland. Saaremaa alone has approximately 2,600 km^2 of land area; the solar installation would occupy less than 5% of this.
- **Northern coastal strip (Pärnu to Paldiski):** Flat agricultural and former industrial land with good grid connection infrastructure and proximity to the planned EstLink3 cable route.
- **Floating offshore solar (experimental component):** Calm Baltic inshore waters offer potential for floating photovoltaic panels that avoid land-use conflicts entirely. This technology is proven in the Netherlands and China and is appropriate for a pilot deployment.

3.1.2. Technical Parameters

Table 1: Solar Array Technical Specifications

Parameter	**Value / Notes**
Installed capacity	10,000–14,000 MW_p
Panel technology	Bifacial monocrystalline silicon (26%+ efficiency, 2026 standard)
Annual yield (Estonia)	950–1,050 kWh/kW_p installed
Summer yield (May–Sep)	Approx. 620–680 kWh/kW_p (60–65% of annual)
Total summer generation	6.2–9.5 TWh (dedicated to hydrogen production)
Land footprint	80–120 km^2 (ground-mounted at 100 MW/km^2)
Capital cost	0.26 million/MW installed (2.6–3.6 billion total)
Operating lifetime	30–35 years
Capacity factor (annual)	11–13%

3.1.3. Why Solar Rather Than Wind for This Application

Wind power, while available year-round, is subject to the well-documented "wind drought" phenomenon in the Nordic-Baltic region: periods of days or weeks in which calm high-pressure systems sit across the entire region simultaneously, causing near-zero wind output across all Baltic and Nordic wind farms at once. This correlated failure is the principal reason hydrogen seasonal storage is necessary.

Solar power, by contrast, has highly *predictable* seasonal output: it is reliable in summer (when hydrogen needs to be produced) and reliably absent in winter (when hydrogen needs to be consumed). This predictability is actually advantageous for planning a seasonal storage system — you know with precision when charging will occur and can size storage accordingly.

3.2. Component 2: Baltic Sea Desalination Plant

3.2.1. Why Desalination?

Electrolysis requires *ultrapure* deionised water, not tap water or river water. Conventional approaches use municipal water treated through reverse osmosis, but for an installation at this scale, a dedicated coastal desalination plant offers several advantages: it uses an effectively unlimited feedstock (the Baltic Sea), it avoids competition with municipal water supply systems, and — crucially — the Baltic Sea is one of the least saline bodies of water on Earth.

3.2.2. Baltic Salinity Advantage

The Baltic Sea has a salinity of approximately 0.5–0.8% in Estonian waters (the Gulf of Finland and northern Baltic Proper), compared to 3.5% for the open ocean. This

means:

- Reverse osmosis membranes operate at **lower pressure** (8–15 bar versus 55–70 bar for ocean water), dramatically reducing energy consumption
- Energy requirement is approximately **1.0–1.5 kWh per cubic metre** versus 3.0–4.5 kWh/m^3 for ocean desalination
- Membrane lifetime is extended by lower operating pressures
- Pre-treatment requirements are reduced (less biofouling, lower turbidity)

3.2.3. Water Requirements and Plant Sizing

Electrolysis consumes approximately **9 litres of pure water per kilogram of hydrogen** produced. For a target output of 400,000 tonnes of hydrogen per summer season:

$$9\,\mathrm{L/kg} \times 400{,}000{,}000\,\mathrm{kg} = 3{,}600{,}000\,\mathrm{m^3} \text{ per season}$$

Distributed across a 150-day summer operating season, this requires a plant capacity of:

$$\frac{3{,}600{,}000}{150} = 24{,}000\,\mathrm{m^3/day}$$

This is a large but entirely standard industrial desalination scale. For comparison, the Sorek B desalination plant in Israel produces 200,000 m^3/day; the proposed Estonian facility is approximately one-eighth that size.

Table 2: Desalination Plant Specifications

Parameter	**Value**
Technology	Multi-stage reverse osmosis (RO) with energy recovery
Capacity	20,000–28,000 m^3/day
Energy consumption	1.0–1.5 kWh/m^3
Total energy demand	20–42 GWh per summer season
Capital cost	60–120 million
Operating cost	8–15 million per year
Brine disposal	Diluted discharge to open Baltic (impact minimal at low input salinity)

3.2.4. In-Channel Micro-Turbines

As water flows from the desalination plant through the purification and distribution pipelines to the electrolyser, controlled pressure drops across purpose-designed

segments of the channel can drive small run-of-river turbines. This is standard practice in water treatment facilities worldwide. At the flow rates required by this system, the recoverable power is estimated at:

$$P = \rho g Q H \eta = 1000 \times 9.81 \times 0.28 \times 12 \times 0.85 \approx \mathbf{28}\,\text{kW per channel segment}$$

Across a full pipeline network of 20–40 such segments, this yields approximately 0.5–1.1 MW of continuous generation during summer operation. This is a relatively small contributor to overall output but represents free electricity with near-zero marginal cost — an efficiency gain worth incorporating into the design.

3.3. Component 3: Industrial Electrolysis Facility

3.3.1. Technology Selection

Two electrolyser technologies are mature enough for deployment at this scale:

Table 3: Electrolyser Technology Comparison

Parameter	**PEM (Proton Exchange Membrane)**	**Alkaline**
Efficiency	65–70%	63–70%
Response time	Seconds (ideal for solar)	Minutes
Capital cost (2026)	700–900/kW	500–700/kW
Lifetime	80,000–100,000 hours	100,000+ hours
Maturity	High; scaling rapidly	Very high; proven technology
Recommended use	Best match for solar variability	Base load, steady operation

Recommendation: A hybrid installation combining PEM electrolysers to handle the variable daily solar profile with alkaline units for steady base operation during peak irradiance hours. This optimises capital cost while maintaining responsiveness to solar intermittency.

3.3.2. Scale and Output

To produce 400,000 tonnes of hydrogen in a 150-day summer season:

$$\text{Daily output target} = \frac{400{,}000{,}000\,\text{kg}}{150\,\text{days}} = 2{,}667{,}000\,\text{kg/day} \tag{1}$$

$$\text{Energy per kg } H_2 \approx 50\,\text{kWh (at 70\% efficiency)} \tag{2}$$

$$\text{Power required} = 2{,}667{,}000 \times 50\,\text{kWh}/24\,\text{h} \approx \mathbf{5{,}556}\,\text{MW continuous} \tag{3}$$

This aligns with the output of the proposed 10,000–14,000 MW solar array when operating at its summer capacity factor of approximately 18–22% over daylight hours, or at an average of roughly 45–55% during peak summer daylight periods.

3.4. Component 4: Geological Hydrogen Storage

3.4.1. The Storage Challenge

Storing 100,000–400,000 tonnes of hydrogen is a significant engineering challenge. Hydrogen has very low volumetric energy density — one tonne of hydrogen occupies approximately 11,000 m^3 at atmospheric pressure, and even at 700 bar compression it occupies around 80 m^3. At the scale required, only geological storage is economically viable.

3.4.2. Estonian Geological Suitability

Estonia's geology offers two credible options:

- **Salt caverns:** Halite (rock salt) deposits exist in the Devonian sedimentary sequence in southern and eastern Estonia. Solution-mined salt caverns are the gold standard for hydrogen storage globally (used extensively in the UK and USA) — impermeable, structurally stable, and re-fillable indefinitely. A geological survey would be required to confirm suitable depth and volume, but preliminary data is promising.
- **Depleted oil shale mines:** The exhausted mines of Ida-Viru county represent an alternative, though these require additional engineering to ensure gas-tightness. They offer the advantage of existing infrastructure, local workforce familiarity, and potential positive narrative around the transformation of a declining industrial region.

Table 4: Hydrogen Storage Specifications

Parameter	**Value**
Storage target	100,000–400,000 tonnes of H_2
Preferred technology	Underground salt caverns
Storage pressure	100–200 bar (cavern-dependent)
Estimated cavern volume required	8–30 million m^3
Capital cost	3–8 billion (dominant cost driver of entire system)
Cycle efficiency	94–97% (minimal losses in geological storage)

3.5. Component 5: Winter Power Generation and Export

3.5.1. Domestic Electricity Production

During winter months, hydrogen is withdrawn from storage and converted to electricity via one of two pathways:

Pathway A — Hydrogen Combined Cycle Gas Turbines (H_2-CCGT): Modified natural gas turbines capable of running on 100% hydrogen are commercially available from Siemens, Mitsubishi and GE as of 2025. Efficiency approximately 55–60%. Capital cost approximately 0.8–1.0 million per MW installed. Best for large baseload generation.

Pathway B — Hydrogen Fuel Cells (PEMFC or SOFC): Higher efficiency (55–65%) but higher capital cost (2–4 million/MW). Best for flexible, dispatchable generation that can respond quickly to grid frequency signals. Produces heat as a byproduct that can be fed into Tallinn's district heating network.

Recommended approach: A combination, with H_2-CCGT providing bulk baseload winter generation and fuel cell installations in or near Tallinn providing both electricity and district heating, maximising total energy recovery.

3.5.2. Round-Trip Efficiency Analysis

It is important to be transparent about the energy losses inherent in this system:

$$\begin{aligned}
\eta_{\text{solar}} &= 22\% \quad \text{(panels to DC electricity)} \\
\eta_{\text{inverter}} &= 97\% \quad \text{(DC to AC)} \\
\eta_{\text{electrolysis}} &= 70\% \quad \text{(electricity to } H_2\text{)} \\
\eta_{\text{storage}} &= 96\% \quad \text{(storage losses over 6 months)} \\
\eta_{\text{reconversion}} &= 58\% \quad (H_2 \text{ turbine to electricity}) \\
\eta_{\text{total}} &= 0.22 \times 0.97 \times 0.70 \times 0.96 \times 0.58 \approx \mathbf{8.5}\% \qquad (4)
\end{aligned}$$

This means that for every 100 units of solar energy captured, approximately 8.5 units reach the Estonian grid as winter electricity. This sounds low — and by the standards of a pumped hydro system (75–80% round-trip) it is. However, the critical insight is that solar energy in summer is effectively free at the margin: once the panels are installed, the electricity they produce costs almost nothing. The relevant comparison is not "solar hydrogen vs pumped hydro" but "solar hydrogen vs paying 94.5/MWh to import winter electricity from Finland." On that comparison, the economics are significantly more favourable.

3.5.3. EU Hydrogen Export

The more economically powerful use of the stored hydrogen is *direct export* without reconversion. Industrial hydrogen commands 3–6/kg on the EU market, equivalent to 90–180/MWh of energy content. This is comparable to or better than Estonian electricity import prices, achieved without the efficiency loss of reconversion.

Estonia is geographically well positioned for this:

- The EU's planned **European Hydrogen Backbone** pipeline network includes a Baltic corridor connecting Finland, Estonia, Latvia, Lithuania and Poland to the German and Benelux industrial heartland
- Finland is an immediate natural customer: Finnish steel, paper and chemical industries are under regulatory pressure to decarbonise and are actively seeking green hydrogen supply
- The proposed **EstLink3** cable (under planning, Eesti osa 50%, 700 MW) provides a template for the bilateral energy infrastructure agreements that would also underpin a hydrogen pipeline
- Germany has declared green hydrogen imports a strategic national priority and has signed preliminary agreements with twelve countries; Estonia is not yet among them but could be

4. Financial Analysis

4.1. Capital Cost Breakdown

Table 5: EIGHES Capital Investment Requirements

#	Component	Low (bn)	High (bn)
1	Coastal solar array (10,000–14,000 MW at 0.26m/MW)	2.6	3.6
2	PEM & alkaline electrolysis facility	2.0	4.0
3	Underground H_2 storage (geological caverns)	3.0	8.0
4	Baltic desalination plant (20,000–28,000 m^3/day)	0.06	0.12
5	H_2 turbines and fuel cells (500–800 MW)	0.5	1.5
6	Grid connection and transmission upgrades	0.3	0.6
7	Pipeline infrastructure (to Finnish/Latvian border)	0.2	0.5
8	In-channel micro-turbines	0.01	0.03
9	Engineering, procurement, construction management	0.4	0.9
10	Contingency (15%)	1.3	2.9
	Total Capital Investment	**10.4**	**22.2**

Note on Storage Cost Uncertainty

Underground geological storage represents the single largest uncertainty in this cost estimate. The actual cost depends critically on the geology — salt cavern storage could be at the low end if suitable formations are confirmed in Estonian territory, while surface pressurised tank storage would be at the high end. An immediate priority for advancing this proposal is a dedicated geological survey of Estonia's Devonian salt formations and exhausted mine workings.

4.2. Operating Costs

Annual operating and maintenance costs once fully constructed:

Table 6: Annual Operating Cost Estimates

Component	**Low (m/year)**	**High (m/year)**
Solar array O&M (1.5% of capex)	39	54
Electrolyser maintenance (3% of capex)	60	120
Storage maintenance	30	80
Desalination operations	8	15
Power generation O&M	15	40
Staff (est. 800–1,200 permanent positions)	40	90
Insurance and administration	20	50
Total Annual OpEx	**212**	**449**

4.3. Revenue Projections

Table 7: Annual Revenue Projections

Stream	**Basis**	**Low (m)**	**High (m)**
H_2 export	200,000–350,000 t at 3.5–5.0/kg	700	1,750
Winter electricity	1.5–2.0 TWh at 110–140/MWh	165	280
District heating (Tallinn)	Fuel cell waste heat	20	60
Micro-turbine generation	0.5–1.1 MW continuous (summer)	2	5
EU carbon credits (avoided imports)	Est. 50,000–100,000 t CO_2 avoided	5	15
Total Annual Revenue		**892**	**2,110**
Net Annual Return (Rev - OpEx)		**680**	**1,661**

4.4. Return on Investment and Payback Period

Table 8: Financial Return Summary (midpoint estimates)

Metric	Estimate
Total capital cost (midpoint)	16 billion
EU grant / co-financing (est. 20–30% of eligible capex)	2–4 billion
Net private/state investment required	12–14 billion
Annual net return (midpoint)	1.17 billion
Simple payback period (without subsidies)	14–20 years
Simple payback period (with EU co-financing)	11–15 years
25-year NPV (8% discount rate)	2–6 billion positive
IRR (estimated range)	6–12%

Comparison to ENMAK 2035

For context, Estonia's existing ENMAK 2035 national energy plan already envisages total investment of 5.6–10 billion in renewable energy infrastructure through 2035. The EIGHES proposal essentially *doubles down* on that commitment, but with a coherent storage and export strategy that transforms the economics from a cost centre into a revenue-generating national asset. The proposal is not an alternative to ENMAK 2035 — it is a strategic extension of it.

5. EU Funding and Policy Alignment

5.1. Available Funding Mechanisms

The EIGHES proposal is exceptionally well-aligned with current EU funding priorities. The following mechanisms are directly applicable:

- **EU Innovation Fund (Phase 2, 2026–2030):** Dedicated to large-scale clean energy innovation. Green hydrogen production facilities are an explicit priority. Grants of up to 60% of capital cost are available for qualifying projects. The EIGHES electrolyser and storage components would be competitive candidates.
- **REPowerEU:** The European Commission has allocated over 200 billion to accelerate the green transition following the 2022 energy crisis. Green hydrogen from EU member states is a top priority. Estonia's proposal to become a hydrogen exporter within the EU hydrogen backbone network is precisely the kind of project REPowerEU was designed to fund.
- **Connecting Europe Facility (CEF) Energy:** Funds trans-European energy infrastructure including hydrogen pipelines and electrolyser hubs. A Baltic hydrogen corridor connecting EIGHES to Finland and Latvia would be eligible.

- **European Investment Bank (EIB) Green Bonds:** The EIB has a specific programme for green hydrogen infrastructure and is actively seeking Baltic region projects. Long-term low-interest financing (20–30 year terms) at approximately 3–4% is available.
- **Just Transition Fund:** Ida-Viru county, the heartland of Estonian oil shale extraction, qualifies for Just Transition funding as a region in structural economic decline due to fossil fuel phase-out. A significant portion of EIGHES physical infrastructure could be located in Ida-Viru, attracting additional EU support and providing political legitimacy.

5.2. Policy Alignment

Table 9: EIGHES Alignment with Key Policy Frameworks

Policy Framework	How EIGHES Aligns
EU Green Deal	Directly enables clean energy transition; replaces fossil fuel imports
Fit for 55	Displaces 50,000–100,000 t CO_2/year of imported electricity
REPowerEU	Green hydrogen production and export; reduces Russian energy dependency
ENMAK 2035	Provides the seasonal storage missing from current national plan
Baltic Energy Market Interconnection Plan	Adds new energy export revenue stream
EU Hydrogen Strategy	Establishes Estonia as a green hydrogen supplier nation
Just Transition Fund	Regenerates Ida-Viru through new industrial activity
NATO Energy Security	Reduces Baltic energy vulnerability to geopolitical disruption

6. Implementation Roadmap

6.1. Phased Development Plan

The scale of EIGHES requires a phased approach spanning approximately 12–15 years from inception to full operation. The following roadmap is proposed:

Table 10: EIGHES Implementation Phases

Phase	Activities	Timeline
Phase 0 Pre-Development	Geological survey (salt caverns & mines); feasibility study; environmental impact assessment; EU funding applications; site selection; stakeholder consultation	2026–2027
Phase 1 Pilot	Pilot installation: 500 MW solar + small-scale electrolyser (10 MW) + demonstration desalination unit; prove integration; validate storage geology; negotiate H_2 offtake agreements	2027–2029
Phase 2 Scale-Up	Scale to 3,000 MW solar; 500 MW electrolyser; first commercial hydrogen sales; commission initial cavern storage; connect to national grid for winter discharge	2029–2032
Phase 3 Full Build	Full 10,000–14,000 MW solar; complete electrolyser array; full storage capacity; pipeline connection to Finnish/Latvian hydrogen network; full export operations	2032–2036
Phase 4 Maturity	System optimisation; expand export partnerships; explore offshore floating solar extension; consider second electrolyser site in Ida-Viru	2036+

6.2. Critical Path Items

The following items are on the critical path and should be initiated immediately:

1. **Geological survey** of Estonian salt formations — this single item determines the viability and cost of the entire storage component
2. **EU funding application** to the Innovation Fund Phase 2 — application windows are competitive and preparation takes 12–18 months
3. **Offtake agreements** with Finnish and German industrial hydrogen buyers — these de-risk the project commercially and are prerequisite for private financing

4. **Grid impact study** by Elering AS — to determine what grid reinforcements are needed to integrate 10,000+ MW of new solar generation
5. **Planning and environmental permitting** for coastal solar sites — timeline typically 3–5 years in Estonia for installations of this scale

7. Risk Analysis

Table 11: Risk Register

Risk	Description	Rating	Mitigation
Geological storage unavailability	Estonian salt formations may be insufficient	High	Survey immediately; design surface storage fallback
Hydrogen price decline	EU H_2 prices may fall as supply expands	Medium	Long-term offtake contracts; diversify to electricity
Technology cost overrun	Electrolyser costs higher than projected	Medium	Phased construction; index to market
EU policy reversal	Hydrogen support framework changes	Low	Diversified revenue streams
Grid integration challenges	Elering unable to absorb 10+ GW new solar	Medium	Phase construction; build grid in parallel
Water quality issues	Baltic salinity/pollution variability	Low	Multi-stage pretreatment; water quality monitoring
Public opposition	Coastal development on Saaremaa/Hiiumaa	Medium	Early consultation; Just Transition framing in Ida-Viru
Financing gap	Private capital reluctant at this scale	Medium	EU grant first; phase to demonstrate returns

8. Economic and Social Impact

8.1. Macroeconomic Impact

At full operation, EIGHES would represent a transformative addition to the Estonian economy:

- **Export revenue:** Annual hydrogen export revenues of 700 million to 1.75 billion would represent 2–5% of Estonia's current GDP (38 billion, 2024) from a single infrastructure system — an extraordinary return for a country of 1.3 million people
- **Import substitution:** Replacing 512 million/year of electricity imports reduces the current account deficit and retains capital within the Estonian economy
- **Energy price stability:** Domestic hydrogen-backed electricity generation would be immune to the Nord Pool price spikes that have driven consumer price increases, providing a price ceiling for winter electricity
- **Industrial attraction:** Green hydrogen availability at competitive prices would attract energy-intensive green industries (green steel, ammonia, e-fuels) seeking to decarbonise

8.2. Regional Development: Ida-Viru

The northeastern county of Ida-Viru faces the most acute structural adjustment of any Estonian region as oil shale closes. Unemployment will rise substantially as the Narva plants and associated mining operations wind down through 2035. EIGHES presents an opportunity to partially redirect the workforce and industrial base:

- Exhausted mine workings potentially repurposed as hydrogen storage infrastructure
- Existing high-voltage transmission infrastructure and grid connections retained and repurposed
- An industrial workforce with relevant skills in large-scale energy plant operations directly transferable to hydrogen facility management
- EU Just Transition funding specifically targeted at this region

8.3. Employment

Table 12: Employment Estimates

Employment Category	Jobs (Construction)	Jobs (Operations)
Solar array installation	6,000–10,000	200–400
Electrolyser facility	1,000–2,000	200–400
Storage construction	500–1,500	100–200
Desalination plant	200–400	50–100
Power generation	300–600	150–300
Pipeline and grid	500–1,000	100–200
Indirect/supply chain	2,000–5,000	500–1,000
Total	**10,500–20,500**	**1,300–2,600**

9. Conclusions and Recommended Next Steps

9.1. Summary of the Case

Estonia faces a genuine and urgent energy security challenge. The closure of oil shale generation, while environmentally necessary, creates a structural winter electricity deficit that current planning — focused on adding renewable generation without addressing seasonal storage — does not solve. The result, without intervention, is a deepening dependency on expensive electricity imports, rising consumer prices, and an increasingly vulnerable grid in a geopolitically sensitive region.

The Estonia Integrated Green Hydrogen Energy System proposes a coherent, technically grounded and financially credible response. By converting Estonia's abundant summer solar resource into storable, exportable hydrogen, the system simultaneously:

- Provides the seasonal storage that ENMAK 2035 currently lacks
- Generates 0.7–1.75 billion in annual export revenue from a new internationally traded commodity
- Reduces electricity import dependency and insulates consumers from Nord Pool price volatility
- Positions Estonia as a first-mover green hydrogen exporter within the EU
- Supports the economic transition of Ida-Viru county
- Is fully aligned with every major EU energy and climate policy framework

The capital requirement of 10–22 billion is substantial but comparable to infrastructure investments Estonia has already committed to under ENMAK 2035, and is well within the range of EU-co-financed national energy transformation projects being

undertaken across the member states.

9.2. Recommended Immediate Actions

Recommended Next Steps (2026–2027)

1. Commission a **detailed geological survey** of Estonian Devonian salt formations and exhausted oil shale mine workings for hydrogen storage potential
2. Engage **Elering AS** and the Estonian Ministry of Climate to initiate a formal grid impact assessment for 10,000+ MW additional solar generation
3. Submit a **Concepts Note** to the EU Innovation Fund Phase 2 secretariat to test eligibility and receive preliminary feedback
4. Initiate **diplomatic conversations** with Finnish and German energy ministries regarding potential hydrogen offtake frameworks
5. Commission a **full independent feasibility study** by a qualified energy engineering consultancy, to produce bankable technical and financial analysis
6. Establish a **cross-ministerial working group** (Climate, Economy, Finance, Foreign Affairs) to coordinate the proposal's development

The energy transition is not simply a challenge to be managed — it is, for the right countries in the right position with the right resources, an opportunity to be seized. Estonia has the solar resource, the geology, the industrial base, the EU membership, and the geopolitical motivation to become a green hydrogen exporting nation. The only thing required is the decision to pursue it.

Prepared by Jordan Thomas Brodie, March 2026

All enquiries and correspondence regarding this proposal are welcome.

Appendix A: Key Technical Assumptions

Table 13: Technical Assumptions Used in Calculations

Parameter	**Value**	**Source basis**
Solar irradiance, Estonia	950–1,050 kWh/kW_p/year	PVGIS 2024
Panel efficiency	22% (bifacial mono-Si)	Industry standard 2026
Electrolysis efficiency	70% (LHV basis)	IEA 2024
Baltic salinity	0.5–0.8%	HELCOM monitoring data
Desalination energy	1.0–1.5 kWh/m^3	Low-salinity RO benchmark
Water per kg H_2	9 litres (pure; approx 18 litres raw)	NREL
H_2 energy content	33.3 kWh/kg (LHV), 39.4 kWh/kg (HHV)	NIST
Storage round-trip efficiency	94–97%	Underground cavern literature
Turbine efficiency (H_2-CCGT)	55–60%	Siemens / GE datasheets 2025
Fuel cell efficiency (PEMFC)	60–65%	DOE 2025
EU H_2 market price	3.0–6.0/kg (green H_2, 2026)	BloombergNEF
Solar capex	0.26 million/MW	ENMAK 2035 reference
Estonia winter electricity gap	1.5–2.0 TWh	Elering VKA 2024

Appendix B: Glossary

EIGHES Estonia Integrated Green Hydrogen Energy System (this proposal)

ENMAK *Energiamõjude kava* — Estonia's National Energy Development Plan

PEM Proton Exchange Membrane — a type of electrolyser/fuel cell technology

CCGT Combined Cycle Gas Turbine — high-efficiency power generation

RO Reverse Osmosis — the primary technology for water desalination

LCOE Levelised Cost of Electricity — lifetime cost per unit of energy produced

LHV Lower Heating Value — the usable energy content of hydrogen (33.3 kWh/kg)

ETS Emissions Trading System — the EU carbon market

REPowerEU European Commission programme to accelerate the clean energy transition

CEF Connecting Europe Facility — EU infrastructure funding programme

P2X Power-to-X — conversion of electricity into chemical energy carriers (H_2, ammonia, methanol)

Nord Pool The pan-Nordic and Baltic electricity exchange market

Elering Estonian electricity and gas transmission system operator

HELCOM Baltic Marine Environment Protection Commission